

Simplefind program Screenshots:

For reference this is what is included in /testdir

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ls -R testdir
testdir:
d1 d2 f1.c f1.txt

testdir/d1:
f2.c

testdir/d2:
f2.c f3.c f4.c link1
```

Standard:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind
.
./simplefind
./testdir
./testdir/d2
./testdir/d2/f4.c
./testdir/d2/.hidden.c
./testdir/d2/link1
./testdir/d2/f2.c
./testdir/d2/f3.c
./testdir/f1.c
./testdir/f1.txt
./testdir/d1
./testdir/d1/f2.c
./simplefind.o
./Makefile
./simplefind.c
```

Standard with defined starting path:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind testdir
testdir
testdir/d2
testdir/d2/f4.c
testdir/d2/.hidden.c
testdir/d2/link1
testdir/d2/f2.c
testdir/d2/f3.c
testdir/f1.c
testdir/f1.txt
testdir/d1
testdir/d1/f2.c
```

Verbose Standard with defined starting path (symlink included):

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -l testdir
69725 4 drwxr-xr-x 3 tunabeluga tunabeluga 4096 Oct  2 23:32 .
239794 24 -rwxr-xr-x 1 tunabeluga tunabeluga 24000 Oct  2 23:32 ./simplefind
234002 4 drwxr-xr-x 4 tunabeluga tunabeluga 4096 Oct  2 23:42 ./testdir
234004 4 drwxr-xr-x 2 tunabeluga tunabeluga 4096 Oct  2 23:42 ./testdir/d2
239807 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 ./testdir/d2/f4.c
234013 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  1 23:07 ./testdir/d2/.hidden.c
234011 0 lrwxrwxrwx 1 tunabeluga tunabeluga 9 Oct  1 23:07 ./testdir/d2/link1 -> ../f1.txt
14161 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 ./testdir/d2/f2.c
239801 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 ./testdir/d2/f3.c
239800 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 ./testdir/f1.c
234006 4 -rw-r--r-- 1 tunabeluga tunabeluga 6 Oct  1 23:06 ./testdir/f1.txt
234003 4 drwxr-xr-x 2 tunabeluga tunabeluga 4096 Oct  1 23:06 ./testdir/d1
234007 4 -rw-r--r-- 1 tunabeluga tunabeluga 6 Oct  1 23:06 ./testdir/d1/f2.c
239795 20 -rw-r--r-- 1 tunabeluga tunabeluga 20232 Oct  2 23:32 ./simplefind.o
233995 4 -rw-r--r-- 1 tunabeluga tunabeluga 224 Oct  2 23:32 ./Makefile
27512 8 -rw-r--r-- 1 tunabeluga tunabeluga 6428 Oct  2 23:34 ./simplefind.c
```

Pattern with defined starting path:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -n '*.c' testdir
testdir/d2/f4.c
testdir/d2/.hidden.c
testdir/d2/f2.c
testdir/d2/f3.c
testdir/f1.c
testdir/d1/f2.c
```

Verbose Pattern with defined starting path:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -l -n "*.c" testdir
239807 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f4.c
234013 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  1 23:07 testdir/d2/.hidden.c
14161 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f2.c
239801 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f3.c
239800 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/f1.c
234007 4 -rw-r--r-- 1 tunabeluga tunabeluga 6 Oct  1 23:06 testdir/d1/f2.c
```

All the flags included!

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -l -x -n "*.c" testdir
239807 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f4.c
234013 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  1 23:07 testdir/d2/.hidden.c
14161 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f2.c
239801 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/d2/f3.c
239800 0 -rw-r--r-- 1 tunabeluga tunabeluga 0 Oct  2 23:42 testdir/f1.c
234007 4 -rw-r--r-- 1 tunabeluga tunabeluga 6 Oct  1 23:06 testdir/d1/f2.c
```

Testing against Find:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -n "*.c" testdir > sf_out.txt
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ find testdir -name "*.c" > find_out.txt
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ diff sf_out.txt find_out.txt
○ (base) tunabeluga@AndrewGaming:~/ECE357/ps2$
```

(W/ the -l flag, it works but diff fails because of formatting issues)

Device Numbers (Snippet):

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -l /dev
1 0 drwxr-xr-x 15 root root 3840 Sep 15 00:50 /dev
381 0 crw-rw---- 1 root tty 7,133 Sep 15 00:50 /dev/vcsa5
380 0 crw-rw---- 1 root tty 7,69 Sep 15 00:50 /dev/vcsu5
378 0 crw-rw---- 1 root tty 7,5 Sep 15 00:50 /dev/vcs5
374 0 crw-rw---- 1 root tty 7,132 Sep 15 00:50 /dev/vcsa4
373 0 crw-rw---- 1 root tty 7,68 Sep 15 00:50 /dev/vcsu4
372 0 crw-rw---- 1 root tty 7,4 Sep 15 00:50 /dev/vcs4
370 0 crw-rw---- 1 root tty 7,131 Sep 15 00:50 /dev/vcsa3
369 0 crw-rw---- 1 root tty 7,67 Sep 15 00:50 /dev/vcsu3
367 0 crw-rw---- 1 root tty 7,3 Sep 15 00:50 /dev/vcs3
363 0 crw-rw---- 1 root tty 7,130 Sep 15 00:50 /dev/vcsa2
362 0 crw-rw---- 1 root tty 7,66 Sep 15 00:50 /dev/vcsu2
361 0 crw-rw---- 1 root tty 7,2 Sep 15 00:50 /dev/vcs2
360 0 crw-rw---- 1 root tty 7,134 Sep 15 00:50 /dev/vcsa6
359 0 crw-rw---- 1 root tty 7,70 Sep 15 00:50 /dev/vcsu6
358 0 crw-rw---- 1 root tty 7,6 Sep 15 00:50 /dev/vcs6
319 0 drwxr-xr-x 5 root root 100 Sep 15 00:50 /dev/disk
351 0 drwxr-xr-x 2 root root 80 Sep 15 00:50 /dev/disk/by-uuid
354 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-uuid/e39e8cb5-4c41-476e-89b4-51c8b588f14d -> ../../sdc
352 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-uuid/53227d43-b526-4f93-b3a9-e94f25ca189e -> ../../sdd
345 0 drwxr-xr-x 2 root root 120 Sep 15 00:50 /dev/disk/by-path
356 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-path/acpi-VMBUS:00-vmbus-fd1d2cbdce7c535c966beb5f811c95f0-lun-0 -> ../../sda
349 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-path/acpi-VMBUS:00-vmbus-fd1d2cbdce7c535c966beb5f811c95f0-lun-1 -> ../../sdb
347 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-path/acpi-VMBUS:00-vmbus-fd1d2cbdce7c535c966beb5f811c95f0-lun-2 -> ../../sdc
346 0 lrwxrwxrwx 1 root root 9 Sep 15 00:50 /dev/disk/by-path/acpi-VMBUS:00-vmbus-fd1d2cbdce7c535c966beb5f811c95f0-lun-3 -> ../../sdd
320 0 drwxr-xr-x 2 root root 200 Sep 15 00:50 /dev/disk/by-id
```

Error Checking Cases:

Missing pattern to -n:

```
⊗ (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -n
./simplefind: option requires an argument -- 'n'
To run: ./simplefind [-l] [-x] [-n pattern] [starting_path]
```

No Permission to open certain directory:

```
● (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind /root
/root
Cannot open directory /root, skipping! Permission denied
```

Invalid option:

```
⊗ (base) tunabeluga@AndrewGaming:~/ECE357/ps2$ ./simplefind -a
./simplefind: invalid option -- 'a'
To run: ./simplefind [-l] [-x] [-n pattern] [starting_path]
```